Results outline

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Research questions

1. Does tree growth increase with longer growing seasons?
2. Did an earlier start and later end of season increase growth?
3. Does tree growth increase with longer growing seasons and is it consistent across species and provenances?
4. How did growth change across years, and how did it relate to climate?

Results

1. Summary basic information
   (a) Wildchrokie includes four deciduous tree species from four sites with leaf phenology data spaning three years (2018 to 2020) for a total of 239 leafout and 230 budset observations, yielding 227 complete growing seasons (GS) for 81 individuals.
   (b) The calendar and thermal seasons ranged from 90 days and 1465.1 GDD (*B. alleghaniensis* in 2018) to 173 days and 2513.6 GDD (*A. incana* in 2018)
   (c) GS thermal sum ranged from 1465.1 GDD (*B. alleghaniensis* in 2018) to 2513.6 GDD, (*A. incana* in 2018)
   (d) The earliest leafout day of year was 22-Apr (*A. incana* in 2019) and the latest, 05-Jun (also *A. incana*, but in 2020). The earliest budset day of year was 13-Aug (*B. alleghaniensis* in 2018) and the latest, 24-Oct (*A. incana* in 2018)
   (e) Mean summer temperature were similar across years with 2019 being the coolest (22.38°C), 2020 the hottest (23.07°C), and 2018 in between (22.85°C)
   (f) No drought for the three years, except for in 2020, where a moderate summer drought ramped up to a severe drought in September.
   (g) Annual ring widths across years varied from 0.39 mm (*B. alleghaniensis* in 2020) to 13.55 mm (*B. populifolia* in 2018) (Figure S5).
2. Does tree growth increase with longer growing seasons?
   (a) Longer growing seasons increased growth for all species, but varied depending on the metric used to measure it.
   (b) Longer thermal season increased growth for *B. papyrifera* ($\beta = 0.44$, UI_{90} [0.18, 0.7]), *B. populifolia* ($\beta = 0.22$, UI_{90} [0.07, 0.36]), and *B. alleghaniensis* ($\beta = 0.13$, UI_{90} [0.01, 0.24]), but *A. incana* had a weak effect, but trended in the same direction ($\beta = 0.07$, UI_{90} [-0.04, 0.17]) (Figure 1a and e) (TableS1).
   (c) *A. incana* grew more with longer calendar seasons ($\beta = 0.05$, UI_{90} [0, 0.1]), but of the three species that responded to thermal season, only *B. papyrifera* also grew more with longer calendar seasons ($\beta = 0.18$, UI_{90} [0.05, 0.32]).
   (d) *B. alleghaniensis* ($\beta = 0.03$, UI_{90} [-0.02, 0.09]), and *B. populifolia* ($\beta = 0.05$, UI_{90} [-0.02, 0.13]) trended in the same direction (Figure 1b and f) (Table S1).
   (e) Results with basal area increment (BAI) for the GDD model were broadly similar to ring width (see Table S3)
3. Did a later start and end of season increase growth?

(a) A longer calendar season increased growth for two species, but neither grew more with a later start of season.

(b) *A. incana* grew more with longer calendar season, but grew less with a later start of season (β = -0.07, UI_{90} [-0.14, -0.01]) (Figure 1c and g) (Table S1).

(c) *B. alleghaniensis* which did not grow more with longer calendar season, did increase growth with a later start of season (β = 0.1, UI_{90} [0, 0.2]) (Figure 1c and g) (Table S1).

(d) Three species grew more with a later end of season (*B. alleghaniensis*: β = 0.12, UI_{90} [0.05, 0.18], *B. populifolia*: β = 0.1, UI_{90} [0.02, 0.17]), with *B. papyrifera* (β = 0.21, UI_{90} [0.09, 0.33]) being the only one of those three that also grew more with longer calendar season (Figure 1d and h) (Table S1).

4. Does tree growth increase with longer growing seasons and is it consistent across species and provenances?

(a) *B. papyrifera* grew the least (α = -3.82, UI_{90} [-8.73, 1.09]) (Figure S3), but was the most responsive to longer growing seasons (Figure 1a and b).

(b) *A. incana* grew the most (α = 1.01, UI_{90} [-3.14, 5.18]) (Figure S3), but had contrasting responses to longer growing seasons (Figure 1a and b).

(c) Growth did not change across any provenances (Figure 2) (Table S2), and the variation across sites (σ = 0.18, UI_{90} [0.02, 0.5) tree ids (σ = 0.17, UI_{90} [0.05, 0.27]) was similar

5. How did growth change across years, and how did it relate to climate?

(a) Growth was highest in 2018 (0.19, UI_{90} [-0.76, 1.1]) (Figure S5), but that year was not the warmest nor the longest calendar season, unlike our finding that warmer and longer seasons increase growth for most of our species (Figure S1 and Figure S2).

(b) Growth was lowest in 2020 (-0.43, UI_{90} [-1.38, 0.48]) (Figure S5) that year had the warmest thermal, but the shortest calendar season, also unlike our findings that growth increases with warmer seasons, and to a lesser extent, with longer calendar season (Figure S1 and Figure S2).

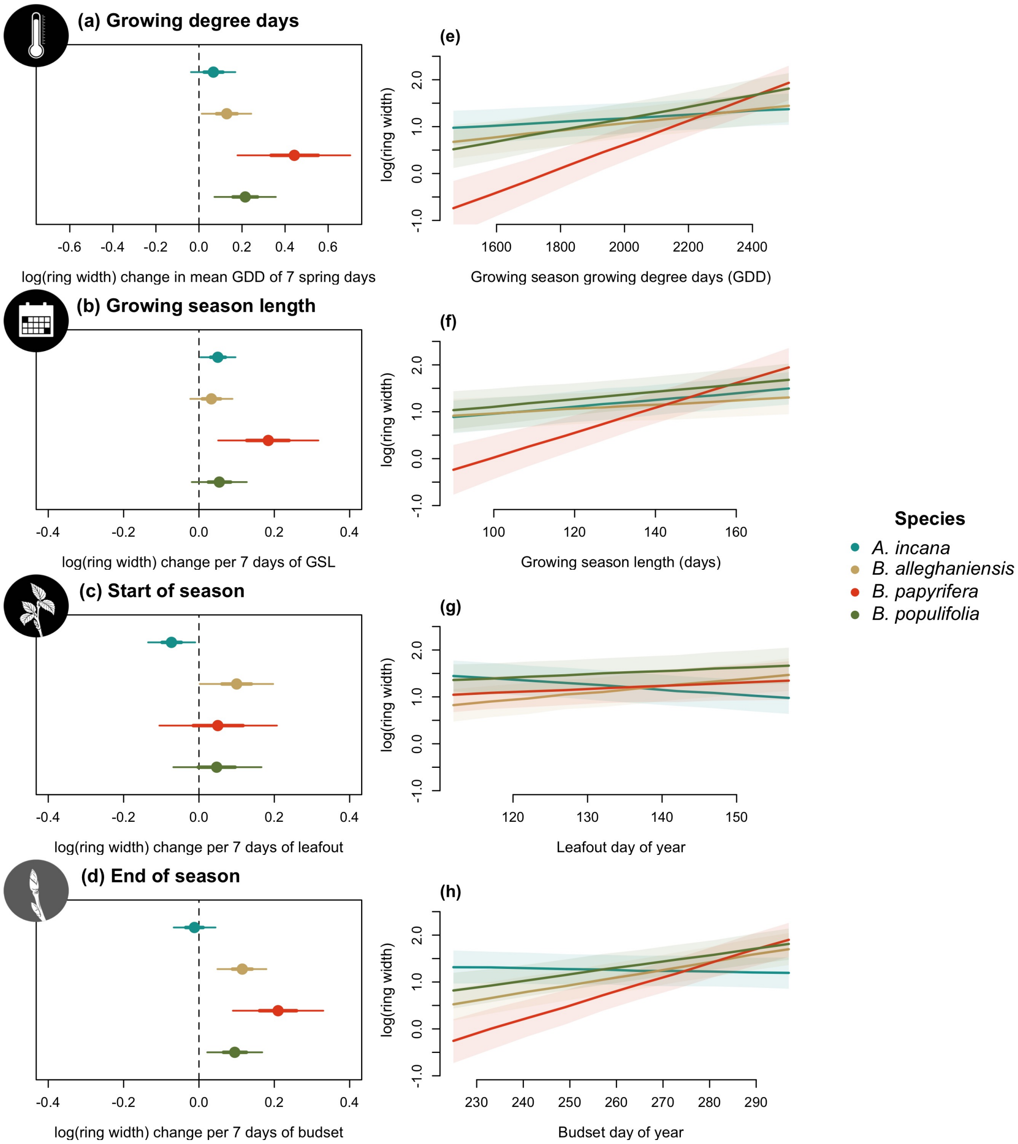

Figure 1: Model slope estimates representing ring width change as a function of four different predictors across four species. GDD represent the accumulated growing degree days between leafout and budset and the models estimates shows ring width change per 7 average spring days GDD (174.05) (a). GSL is the number of days between leafout and budset (b). SOS represent the start of season, which is leafout (c) and EOS, the end of season through budset (d). All predictors were fit using the same model and number of observations, with only the independent variable changing. The dots represent the mean of each species and the lines, the 50% and 90% uncertainty intervals (UI). *A. incana* and *B. papyrifera* grew more with longer calendar seasons, but only *A. incana* grew less with later leafout and had no effect in response to budset. Interestingly, *B. papyrifera* did not grow more with earlier leafout, but did with a later budset and longer calendar season. *B. alleghaniensis* and *B. populifolia* did not grow more with longer calendar seasons, but *B. alleghaniensis* grew more with a later leafout and budset whereas *B. populifolia* only grew more with a later budset.

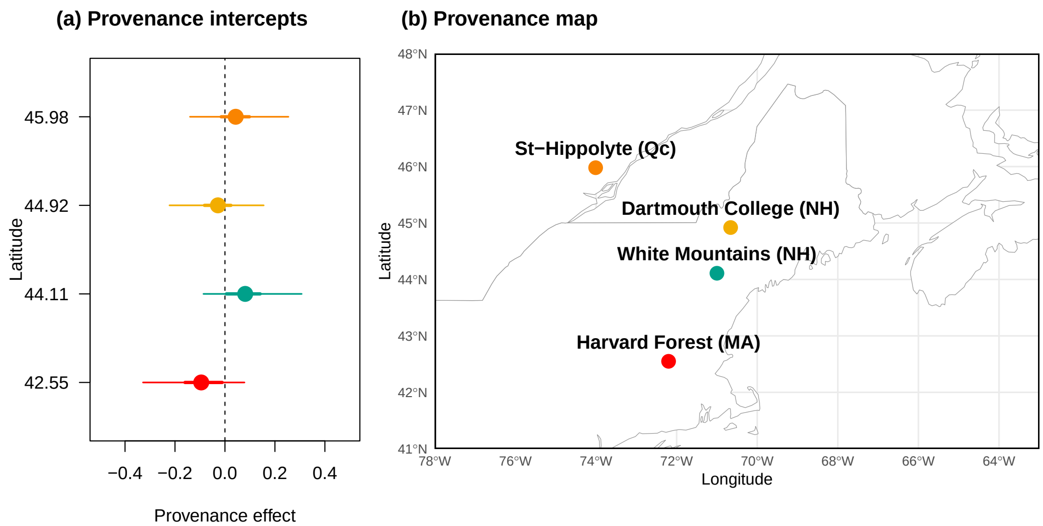

Figure 2: The effect of provenances on growth across a latitudinal gradient. (a) Model intercept parameters for each site with growing degree days (GDD) as the predictor (effects were similar for the other predictors, see Table S2). The dots represent the mean of each provenance and the thick lines, the 50% and 90% uncertainty intervals (UI). The sites are color-coded from (b), the map showing their locations.

Supplemental results

Table S1: Species level slope parameters (mean and 90% uncertainty intervals, in square brackets) across the four different predictors for Wildchrokie. GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season.

Species	GDD	GSL	SOS	EOS
A. incana	0.07 [-0.04, 0.17]	0.05 [0.00, 0.10]	-0.07 [-0.14, -0.01]	-0.01 [-0.07, 0.04]
B. alleghaniensis	0.13 [0.01, 0.24]	0.03 [-0.02, 0.09]	0.10 [0.00, 0.20]	0.12 [0.05, 0.18]
B. papyrifera	0.44 [0.18, 0.70]	0.18 [0.05, 0.32]	0.05 [-0.10, 0.21]	0.21 [0.09, 0.33]
B. populifolia	0.21 [0.07, 0.36]	0.05 [-0.02, 0.13]	0.05 [-0.07, 0.17]	0.10 [0.02, 0.17]

Table S2: Site level intercept parameters (mean and 90% uncertainty intervals, in square brackets) across the four different predictors for Wildchrokie. GDD: growing degree days, GSL: growing season length, SOS: start of season and EOS: end of season.

Site	GDD	GSL	SOS	EOS	Lon.
Harvard Forest (MA)	-0.10 [-0.33, 0.08]	-0.10 [-0.34, 0.08]	-0.05 [-0.26, 0.11]	-0.09 [-0.33, 0.08]	-72.19
White Mountains (NH)	0.08 [-0.09, 0.31]	0.07 [-0.10, 0.29]	0.08 [-0.07, 0.29]	0.08 [-0.10, 0.30]	-71.23
Dartmouth College (NH)	-0.03 [-0.22, 0.15]	-0.05 [-0.26, 0.13]	-0.07 [-0.27, 0.09]	-0.04 [-0.24, 0.15]	-71.15
St-Hippolyte (Qc)	0.04 [-0.14, 0.25]	0.03 [-0.16, 0.24]	0.02 [-0.15, 0.21]	0.05 [-0.14, 0.27]	-74.03

Table S3: Posterior means and 90% uncertainty intervals for all model parameters, when using basal area increment (BAI) as the response variable

Parameter	Estimate
$\sigma_{\alpha_{tree}}$	0.52 [0.41, 0.64]
$\sigma_{\alpha_{site}}$	0.27 [0.03, 0.73]
σ_y	0.63 [0.57, 0.70]
$\beta_{spp_{A.incana}}$	0.11 [-0.03, 0.25]
$\beta_{spp_{B.alleghaniensis}}$	0.20 [0.02, 0.36]
$\beta_{spp_{B.papyrifera}}$	0.92 [0.49, 1.34]
$\beta_{spp_{B.populifolia}}$	0.41 [0.20, 0.61]
$\alpha_{site_{DartmouthCollege(NH)}}$	-0.11 [-0.47, 0.17]
$\alpha_{site_{HarvardForest(MA)}}$	-0.09 [-0.45, 0.19]
$\alpha_{site_{St-Hippolyte(Qc)}}$	0.06 [-0.23, 0.42]
$\alpha_{site_{WhiteMountains(NH)}}$	0.15 [-0.11, 0.52]
$\alpha_{year_{2018}}$	-0.36 [-1.32, 0.61]
$\alpha_{year_{2019}}$	0.33 [-0.65, 1.32]
$\alpha_{year_{2020}}$	-0.04 [-1.01, 0.94]
$\alpha_{spp_{A.incana}}$	3.23 [-2.25, 8.55]
$\alpha_{spp_{B.alleghaniensis}}$	2.09 [-3.40, 7.54]
$\alpha_{spp_{B.papyrifera}}$	-7.59 [-14.84, -0.29]
$\alpha_{spp_{B.populifolia}}$	-0.07 [-5.98, 5.57]

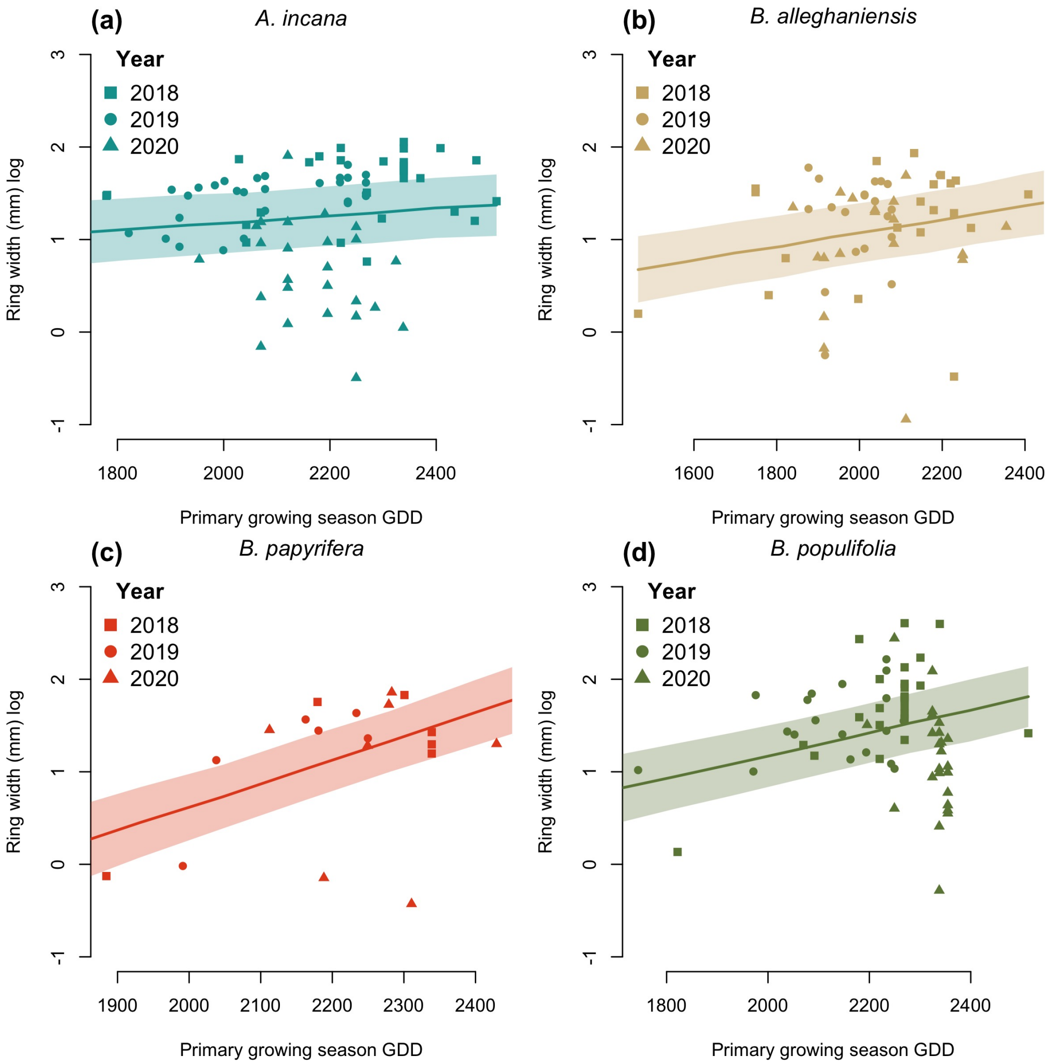

Figure S1: Ring width trends in response to thermal season (GDD). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

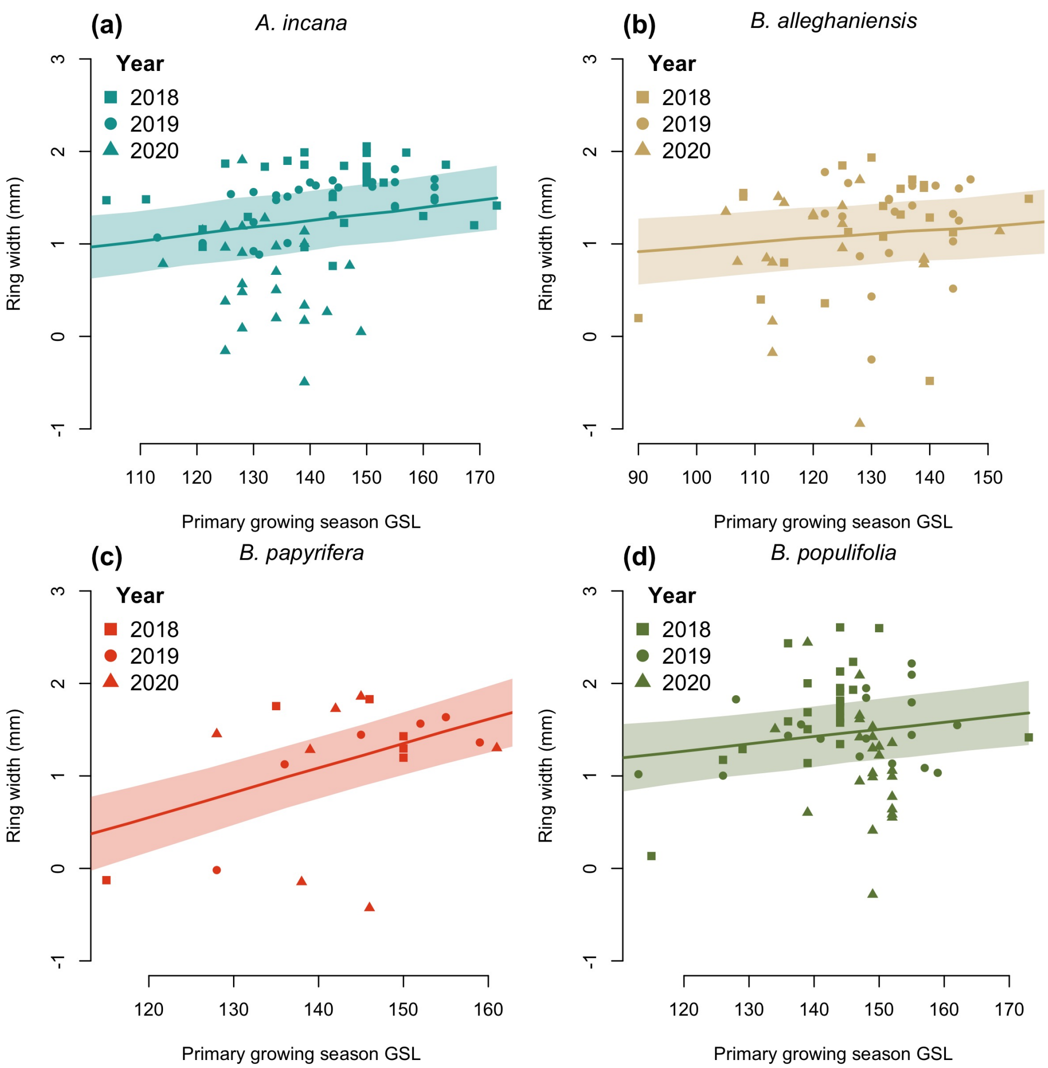

Figure S2: Ring width trends in response to calendar season (GSL). The plots are the reconstructed species estimates posterior distribution with error. The lines represent the mean of the posterior distribution and the shaded area, the 50% UI. The dots are the empirical data shaped by year.

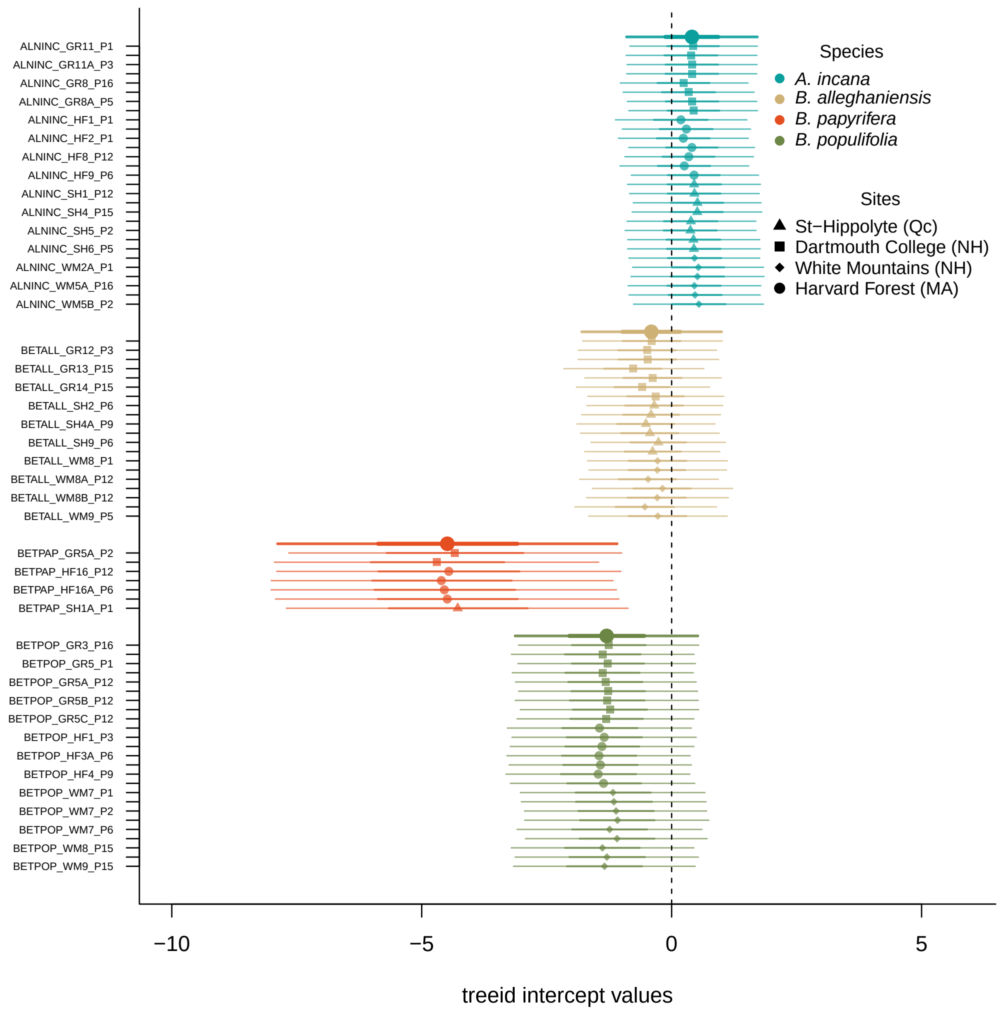

Figure S3: Intercept estimates for each group of the GDD model. The different tree id values represent the sum of the tree id, provenance, species and averaged year intercepts. The larger dots and lines represent aggregated average across these values. The dots are the mean of the posterior distribution and the thick and thin lines, the 50 and 90% UI.

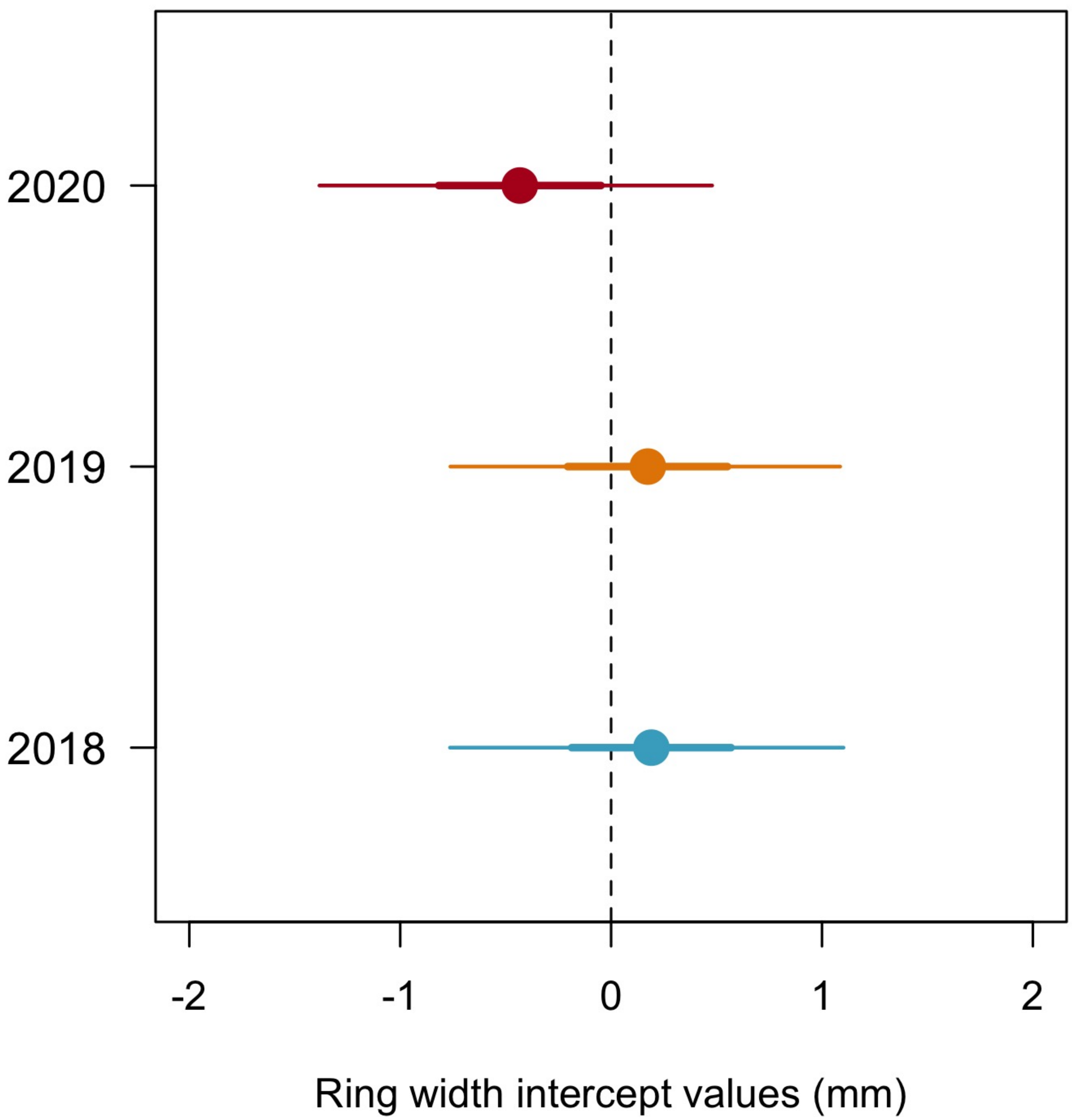

Figure S4: Model estimates for the year intercepts with the dots depicting the mean and the lines, the 50% and 90% uncertainty intervals (UI) and (b) using box plots of the empirical data. Though the trend is weak, 2020 is systematically the year with the lowest growth shown both as the mean of the model intercepts (a) and as the median of the raw data, for each species (b). The growth of 2018 and 2019 is similar.

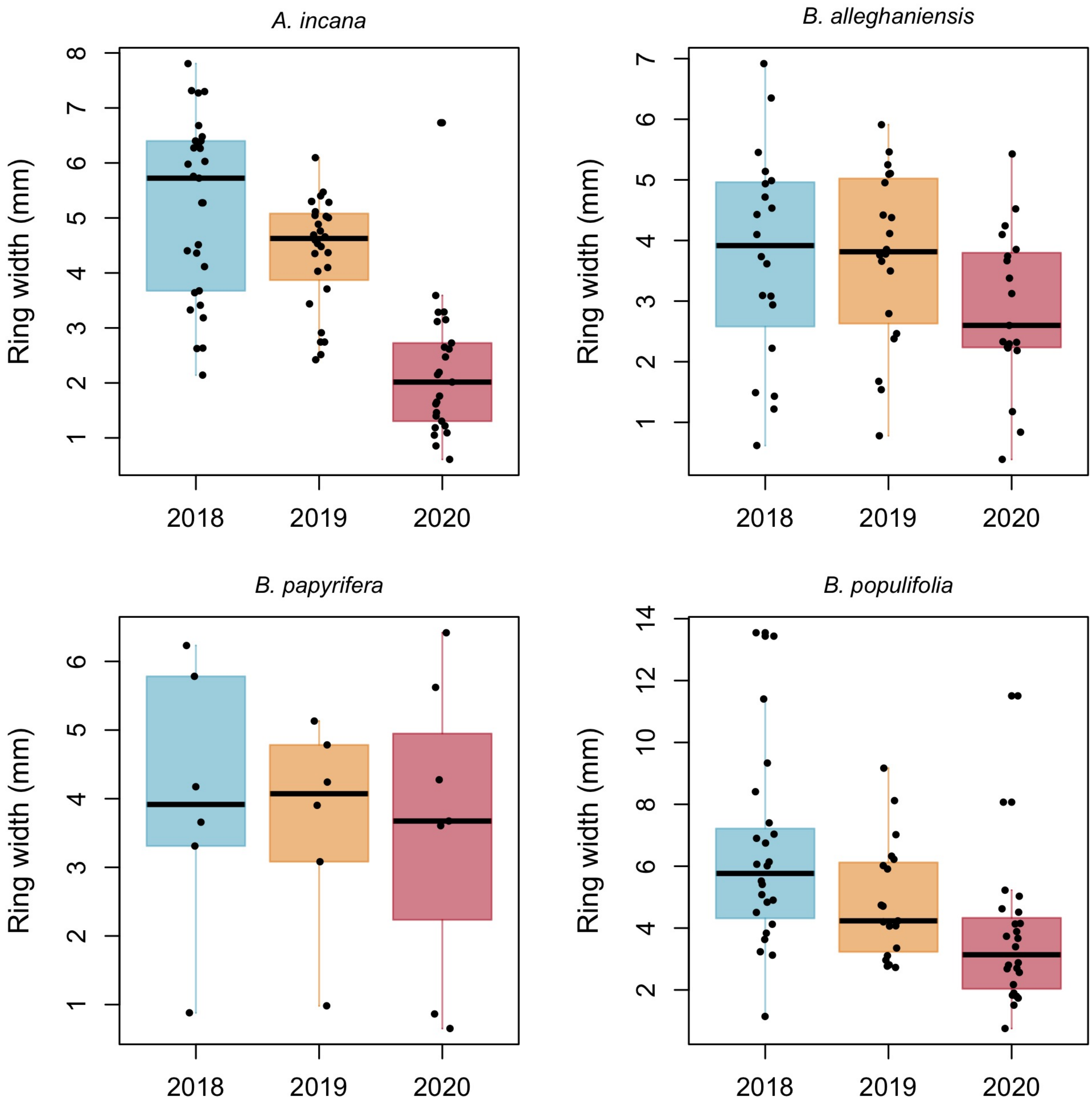

Figure S5: Empirical data. Though the trend is weak, 2020 is systematically the year with the lowest growth shown both as the mean of the model intercepts (a) and as the median of the raw data, for each species (b). The growth of 2018 and 2019 is similar.

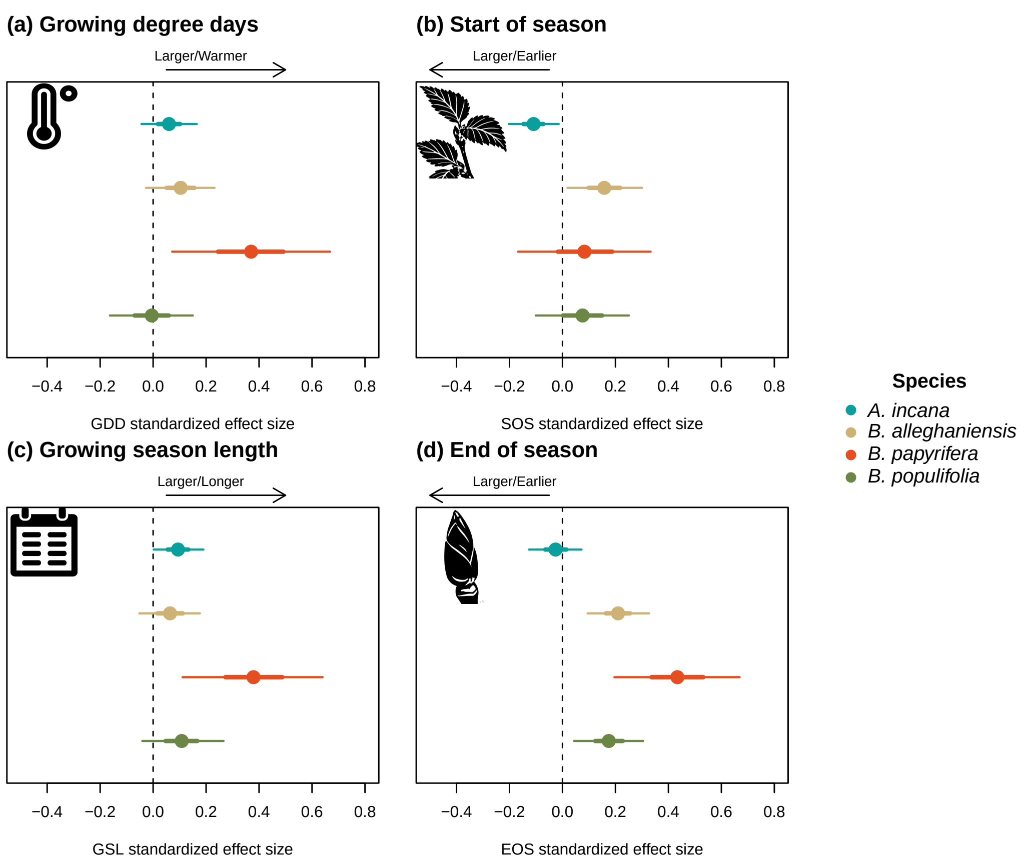

Figure S6: Standardized (z-scored) slope estimates for Wildchrokie. The dots represent the mean of the posterior distribution, the thick lines the 50% uncertainty intervals (UI) and the thinner lines, the 90% UI.